



Laboratory Assessment VI
PEC-2 Laboratory: Full Stack Development - BIT26PE05
To integrate a React frontend application
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AIM:

To integrate a React frontend application with a Spring Boot backend for the Employee Management system and perform CRUD operations through RESTful API calls.

Objectives

- To connect the React frontend with Spring Boot backend using Axios and REST APIs
- To perform CRUD operations (Add, View, Update, Delete) through the user interface
- To ensure proper data flow between frontend, backend, and H2 database
- To implement basic validation and error handling for success and failure cases
- To test full-stack functionality and verify real-time data updates in the UI

Employee Management

Add

Name	Role	Salary	Actions
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CRUD OPERATIONS FROM UI (MANDATORY)

You already built them — now you must **show them clearly**

◆ ADD (POST)

- Enter data in UI
- Click Add
- Data appears in table

Employee Management

Add

Name	Role	Salary	Actions
Ram	Developer	10000	<button>Edit</button> <button>Delete</button>

Employee Management

Name	Role	Salary	Add
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Name	Role	Salary	Actions
Ram	Developer	10000	Edit Delete
Shyam	Tester	20000	Edit Delete

Run Run Selected Auto complete

```
SELECT * FROM EMPLOYEES;
```

```
SELECT * FROM EMPLOYEES;
```

ID	NAME	ROLE	SALARY
1	Ram	Developer	10000.0
2	Shyam	Tester	20000.0

(2 rows, 2 ms)

◆ VIEW (GET)

Employee Management

Name	Role	Salary	Add
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Name	Role	Salary	Actions
Ram	Developer	10000	Edit Delete
Shyam	Tester	20000	Edit Delete

◆ UPDATE (PUT)

- Click Edit
- Modify data
- Click Update
- Data changes in UI

Employee Management

Name	Role	Salary	Actions
Ram	Developer	200000	Edit Delete
Shyam	Tester	20000	Edit Delete

```
SELECT * FROM EMPLOYEES;
```

ID	NAME	ROLE	SALARY
1	Ram	Developer	200000.0
2	Shyam	Tester	20000.0

(2 rows, 11 ms)

◆ DELETE (DELETE)

- Click Delete
- Row removed

Employee Management

Name	Role	Salary	Actions
Ram	Developer	200000	Edit Delete

```
SELECT * FROM EMPLOYEES;
```

ID	NAME	ROLE	SALARY
1	Ram	Developer	200000.0

(1 row, 2 ms)

Conclusion

The assignment successfully demonstrated full-stack integration between the React frontend and Spring Boot backend for the Employee Management system. CRUD operations were performed through the UI using RESTful API calls, and data was dynamically updated and stored in the H2 database. Integration testing, along with basic validation and error handling, ensured reliable and efficient application performance, showcasing complete end-to-end functionality.